Technical Document #410: Software Engineering - Comprehensive Overview

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Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Technical debt represents the cost of choosing easy solutions over better ones.
- Microservices architecture structures applications as independently deployable services.
- Refactoring improves code structure without changing behavior.